

VCU College of Engineering

Project 308

Development of a Customizable Clinical Informatics Web Portal

Preliminary Design

Prepared for

Tianjun Ma

VCU Department of Radiation Oncology

By

Blake Cottrill, Adalberto Gomez, Alycia Logan, and Robert Smith

Under the supervision of

Dr. Thomas Gyeera, Ph.D

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Executive Summary

In collaboration with VCU's School of Health, our team of Computer Science seniors at Virginia Commonwealth University are developing a *Customizable Clinical Informatics Web Portal* (#308) to aggregate multiple fragmented systems in use by the Department of Radiation Oncology in order for a centralized web platform with role-specific workflows and personalized dashboards. Modern radiation oncology requires the use of various independent applications; This project seeks to integrate data from each independent application, utilizing the data to implement a functional database for patient and client record, and normalize them into a unified web application.

This project addresses the problems of inefficient workflows and inconsistent information between the independent systems used by the Department of Radiation Oncology. Our primary goal is to design and implement a web application that takes input from the Department of Radiation Oncology's independent systems (Epic, ARIA, Eclipse, etc), extract the data into a functional backend database while storing it securely, and providing a connected frontend dashboard. The dashboard will be role-based, taking into account the variety of positions found within the Department of Radiation Oncology. Key design requirements include a scalable infrastructure, normalization of data inputted, universal authentication required by the Department of Radiation Oncology, and the use of roles.

Over the course of this semester our team will focus on foundational architectural planning, development of early design diagrams, and researching applicable software tools. Deliverables will include an ERD Diagram establishing relationships between entities within the database, a working prototype of the Radiation Oncology Aggregated Dashboard (ROAD), and the data used in the working database backend. Future deliverables may include a fully functional dashboard with integrated AI for long-term scalability.

This project will combine the team's collaborative research, knowledge of software tools, and their practical implementation in order to optimize the VCU Department of Radiation Oncology's workflow and provide them with an infrastructure with which they can build off of in the future. The Department of Radiation Oncology benefit with a role-based dashboard to assist with work efficiency, while the students benefit by gaining an understanding of industry workload and responsibilities.

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Section A. Problem Statement

The continued exponential growth of software services in our time comes with its own share of cons. Organizations such as the Department of Radiation Oncology here at Virginia Commonwealth University find the need for multiple different applications in order to utilize all the features required for their work. While this approach does work for small scale organizations, fostering a large client base makes the use of these various applications cumbersome. The issue of inconsistent features and information from application to application leads to inefficiencies that weigh down the department's workflow. For an industry as vital as healthcare, relying on fragmented systems can be disastrous, and will most certainly lead to high upkeep costs.

An **Aggregated Dashboard** provides a solution to this problem of fragmentation. The **Radiation Oncology Aggregated Dashboard** (or ROAD), allows for the smaller, fragmented systems in use by the Department of Radiation Oncology to be normalized and sorted cohesively into one contained application. Each fragmented system's data will be ingested into the aggregated dashboard, allowing for consistent access to information and functionality that each system provides. This design offers workflow efficiency, consistency, and scalability for future modification and growth.

The broader engineering goal will be to implement data normalization in a way that balances efficiency and consistency. Reading in information from independent systems while maintaining the integrity of information in order to prevent conflicting data. This brings into question the schema to be implemented. One schema that stands out is **HL7 FHIR**[1] (Fast Healthcare Interoperability Resources), a standard implementation of a medical dashboard, for good reason. FHIR utilizes standard RESTful APIs, allowing for a simplified implementation using HTTP formats and JSON. FHIR uses these formats in order to represent client health information as modular resources [2], a useful design point for normalizing data across differing systems.



Figure 1. Schematic overview of the seven stages of FHIR data modeling [2].

Section B. Engineering Design Requirements

The design requirements for this project are taken from the Department of Radiation Oncology's desired implementation and the academic objectives required by the VCU Computer Science senior capstone project. The aggregated dashboard desired by VCU Health must be open to input data from the existing applications used by the Department of Radiation Oncology, have role-based authorization to access the web application, and be an organized collection of important health data taken from the in-use systems in the Department of Radiation Oncology. The aggregated dashboard aims to enable an efficient workflow for those working in the radiation oncology department and to eliminate inconsistent/redundant information present in the individual systems used by the department.

These requirements will be developed through client discussions, review by the faculty advisor, and comparison to the already existing systems such as Epic, ARIA, and Eclipse.

B.1 Project Goals (i.e. Client Needs)

This project's goals are given by the client from the Department of Radiation Oncology's need for a scalable and practical front-end system. The goals are as follows:

1) Design an architecture that allows for easy input of various data from separate systems into a cohesive application.
2) Normalize given data and provide clear relationships between back-end databases.
3) Provide a practical front-end that has a role-based dashboard and authentication.
4) Allow for future modification in order to build additional functionality.
5) Design for long-term scalability and easy collaboration between users.

B.2 Design Objectives

The key objective of this design is to create a presentable and efficient dashboard. Our main priority is to research the best possible design to determine the best applicable one for our aggregated dashboard. We expect the following from the project:

1) By the end of the academic year, the design will feature a readable and efficient to use interface, displaying specific information pertaining to the user's role.
2) The design will be available, i.e. the radiation oncology department will be able to acquire information from the design whenever needed.
3) The design will provide technical documentation for future users to reproduce and extend the framework whenever necessary.

4) The design can be employed for use in the Department of Radiation Oncology's systems.
5) By the end of Spring 2027, a fully utilizable framework will be delivered.

B.3 Design Specifications and Constraints

These specifications and constraints ensure that ROAD remains secure, scalable, and functional while supporting data integration, role-based access, and the needs of the Department of Radiation Oncology.

Constraints	Type	Objective	Description
Must require authenticated access to ROAD	Security	2, 4	Users must have enterprise authentication before gaining access to ROAD
Must support role-based access	Security and Functional	1, 4	Display information and functions based on the user's assigned role.
Must consolidate patient information.	Functional	1, 4	Provide authorized staff with a merged view of patient data from multiple different sources.
Must support requested clinical dashboard components	Functional	1, 4	The dashboard components are: - Patient list - Treatment progress - Planning - Quality Assurance - Alerts - Timeline - Workflow - System status
Must normalize source data	Functional	4	Convert provided source data into a consistent database structure
Must provide database storage	Functional	2, 4	Store the normalized data in the structured database so that ROAD can retrieve it
Must provide backend API access	Functional	2, 4	Retrieve and format stored data through secured REST API endpoints
Must support user preferences	Usability	1, 3	Store and retrieve individual dashboard and notification preferences

Must support search and filtering	Usability	1	Allow users to search and filter available patient and workflow information
Must maintain system logs	Maintainability	2, 3	Log relevant application activity for monitoring and auditing
Must support future Microsoft Fabric integration	Interoperability	3, 4	Allow database connectivity with Microsoft Fabric
Must deploy to a Windows server	Functional	4,5	Deploy ROAD and required services within the designated Windows server environment
Must operate within project timeline	Schedule	5	Complete ROAD framework within the two-semester capstone schedule.

B.4 Codes and Standards

List all specific codes and standards that are relevant to the design providing specific details of each as they relate to the design. While the terms codes and standards are often used interchangeably, there are in fact important differences in their definitions that should be understood. Standards are documents that provide a set of technical definitions, instructions, rules, guidelines and/or characteristics of a product, process, or service meant to provide consistent and comparable results (e.g. performance requirements, dimensions, testing procedures, file formats etc.). They allow for interchangeability of components and system interoperability and are typically produced by industry or professional organizations such as ASME, ANSI, ASTM, IEEE, ISO, ACM, IAPP, AIS, etc. Standards are meant to help ensure quality, reliability, and safety.

Codes are laws or regulations that specify the methods, materials, components, etc. required for use in a certain product, process, or structure. Codes have been codified into a formal written policy or law and can be approved at the local (municipal), state, or federal level. While standards provide sets of guidelines, codes are constraints that must be met in accordance with the law. It is, however, common for codes to reference or require the use of one or more standards. Some common code producers include the EPA, OSHA, DOTs, and the NFPA. Codes help set minimum acceptable levels in order to protect public health, safety, welfare.

Codes and standards are often listed by their producer followed by an identifying numerical code. They often contain hyphens or periods which may help reference specific parts of a larger code/standard or provide the year of the latest revision. Some general examples in a list of codes and standards are as follows:
ASME Standard No. xxx – design must consider some specific fatigue failure criteria

IEEE Standard No. xxx – design components must not exceed some maximum current limit
ISO Standard No. xxx – design components must adhere to some standard thread size
OSHA Code No. xxx – operators of design must wear appropriate eye and face protection
IRTF Standard No. xxx – design must consider internet communication protocols
W3C Standard No. xxx – design must adhere to some HTML/CSS standards
NIST Standard No. xxx – design must consider some specific data security standards
Note: Relevant codes and standards should be incorporated into the design specifications and constraints listed above.

Section C. Scope of Work

The goal of ROAD is to create a centralized aggregated dashboard that enables the quick and efficient collection and integration of data from various sources, specifically from EPIC, ARIA, and Eclipse. The system will normalize the data to provide a clear representation of the relationships between the underlying backend databases for the front-end dashboard allowing different authentication based on roles.

The project will involve researching and working with FHIR and DICOM interfaces to support data integration. The dashboard will incorporate authentication and authorization through Microsoft Entra ID, OAuth 2.0, and OpenID Connect. Auditing and system monitoring will be supported through health checks, logging, and monitoring tools to make system activity easier to track and audit. The team will provide a readable and efficient interface to use, provide technical documentation for future users to reproduce and extend the framework whenever needed.

C.1 Deliverables

These are the project deliverables that the project team is responsible for providing to the project sponsor. Potential risks that are associated with the project that can damage the future of the project must be identified and discussed with all parties involved.

Deliverables

- Fully designed ERD diagram for creation of database architecture
- Working enterprise authentication architecture with Microsoft Entra ID, OAuth 2.0 / OpenID
- Dashboard that will provide role-based access control for different viewings of dashboard keeping patient information safe
- Unified patient web-portal for displaying all relevant information
- Fully functional REST API back-end
- Personalized dashboard workspace and saved user preference
- Deployed windows server

Potential Risk and Mitigation

- Time issues could arise with not doing appropriate meetings
- Issues with not being able to handle work load
- Not knowing what is to be expected of all team members and not team working

C.2 Milestones

Milestone	Description	Target
Requirements Gathered	Meet with the advisor and Sponsor to identify clinical systems, user roles, data sourcing, security, system architecture.	September 2026
Proposal Approved	Divide the sections and finalize the proposal	October 2026
Develop Environment and and Testing Tools	Set up code repository, hosting access, and testing tools	November 2026
Test Plan Completed	Define test protocols, metrics, tools, and testing schedule	November 2026
Architecture and Design Completed	Select tech stack Deliver system architecture, database schema, and REST API specifications	December 2026
Preliminary Design Document Complete	Report for system architecture and design finalized	December 2026
Authentication and Role-Based Access Prototype	Working SSO login and role-based access with first role specific dashboard	February 2027
Prototype (Data integration)	Connect the first clinical data source through the REST API and display it on the dashboard	March 2027
Security and Functional Testing	Test authentication, access control, and core functions	March 2027
Prototype (Personalized workspace and deployment)	Add workspace customization and deploy to an approved hosting environment	April 2027
Performance and Usability	Run load, response time, and task based test	April 2027

Final Testing	Complete security, performance, usability testing	April 2027
Final Report	Finalized the final report for the presentation	April 2027

C.3 Resources

The project will rely heavily on software and won't require much or any hardware.

Development Resources

- GitHub between team members
- Access to multiple data sources (EPIC, Aria, DICOM metadata) and extracting
- Connection to Microsoft Fabric
- Access to applications for authorization and authentication
- Libraries and External APIs for analysis
- Applications for database creation
- BI and reporting tools

Other Resources

- 4 Team members needed for research and development for ROAD, as well as testing and meeting for project
- Project sponsor providing insight into details of development

Section D. Concept Generation

A number of methods can be used to help generate design concepts from simple reflection and brainstorming, to working the problem backwards, using reverse thinking techniques, and looking to nature for inspiration (i.e. biomimicry). Existing solutions, or components of existing solutions, can be substituted, combined, adapted, modified, put to other uses, eliminated, or rearranged to meet new design objectives and specifications. A minimum of 3 overall design concepts is required for this section although more are welcome. Provide a brief description of how each design concept addresses the design problem. Discuss the potential pros and cons, including and potential risks of failure, of each of these concepts.

It is likely that each design concept may consist of several components. In this case, one or more of these components may offer a sub-problem that can be further explored, modified, or otherwise improved upon. These sub-problems may lead to the addition of several additional

design concepts and may require the inclusion of a design concept chart or matrix to organize all ideas and potential solutions.

Provide any initial design sketches, drawings, 3D renderings, or conceptual models such as dataflow diagrams, process flows, etc. developed during the concept ideation phase. All hand drawings should be drawn to scale using basic engineering drafting tools (i.e. ruler, protractor, and compass). Geometric stencils can also be used to help produce quality hand drawings. Drawings should be presented in a profession manner, preferably done on engineering graph paper and using a high-quality scan. All sketches should be labeled to identify major components and different drawing views or projections if applicable. Basic dimensions should be provided to give a general sense of scale. Label each sketch or drawing with the name of the team member responsible for the sketch, the date it was drawn, and the drawing scale.

Section E. Concept Evaluation and Selection

Using a systematic decision-making process, evaluate each of the design concepts and choose the one that is most likely to succeed in meeting the design objectives and constraints. A Decision Matrix, or Pugh Matrix, helps to analyze alternatives, eliminate biases, and make rational decisions through thought and structure. First, work to develop a set of selection criteria for which to evaluate the previously generated design concepts. Selection criteria often include concepts of performance, cost, safety, reliability, risk, etc. Note that the selection criteria developed here will likely be more general than the project design objectives. As with the design objectives, conversations with the client help define appropriate selection criteria.

In many cases, the client may value the selection criteria differently, preferring that more emphasis be placed on some than others. In this case, weighting factors may be used to place more or less importance on the various criteria in the decision making process. Again, conversations with the client can be used to define criteria weighting factors. Often times, these conversations must be analyzed and interpreted by the team to determine which criteria are more important to the client and by how much. Feel free to discuss the assigned weighting factors with the client to see if they seem accurate.

Next, define an associated metric to represent each criteria. Metrics should be specific and quantifiable, providing numerical values that quantify the often vague concepts of the selection criteria. Metrics can be obtained, generated, or estimated through a number of methods including simple background research, preliminary design calculations, or basic analyses. Note that these metrics do not need to specifically align with the design specifications although there may be some commonality between the two. Provide a brief discussion of the rationale for selecting each of the assigned metrics.

Using the defined metrics, evaluated each design concept against all selection criteria by filling out a Decision Matrix. Design concepts can be compared by using simple rank scoring, raw scoring, or weighted scoring techniques and design concept with which to move forward can be selected. This type of process provides a meaningful, unbiased means for choosing a preliminary design concept prior to moving forward with more comprehensive, detailed analyses as provided in the design methodology section below. The results of this process should be discussed with the project client prior to moving forward with the selected design. Table 1 provides an example of a simple decision matrix.

Table 1. Example of a Decision Matrix.

	Design Concept A	Design Concept B	Design Concept C	Design Concept D
Criteria 1				
Criteria 2				
Criteria 3				
Criteria 4				
Criteria 5				
Total Score				

Note: Weights can be assigned to each criterion if desired.

Section F. Design Methodology

Provide a detailed explanation of the methods that will be used to help evaluate, improve, and evolve the design through the iterative engineering design process. Consider that ultimately, the final design must be verified and validated to ensure that it meets all of the previously developed and listed design objectives and specifications. Verification ensures that the design meets all specification, while validation confirms that the design functions as intended such to meet the client's needs. While it is common for initial design concepts to first be evaluated using simplified design criteria and metrics, the chosen design should be advanced, and later verified, using engineering calculations, computational models, experimental data, and/or testing procedures.

Use this section to describe any underlying physical principles and mathematical equations that govern the design. Provide details of any computer-aided modeling techniques used to evaluate the design including the software used, prescribed boundary conditions, and assumptions. Include a detailed description of any experimental testing methods including required testing equipment, test set-up layout, data acquisition and instrumentation, and testing procedures. If one or more prototypes is to be produced and tested, provide a detailed description of how each will be evaluated.

Note: The contents of this section are expected to vary from project to project. Subsections may be appropriate for providing details of analytical, computational, experimental, and/or testing methods. Some potential subsections that may be included in this section are provided. While critical design equations may be provided here, lengthy mathematical derivations may be included in an appendix. Validation procedures are critical and all projects should address such topic.

F.1 Computational Methods (e.g. FEA or CFD Modeling, example sub-section)

F.2 Experimental Methods (example subsection)

F.3 Architecture/High-level Design (example subsection)

F.5 Validation Procedure

Describe how the design team will validate that the final design meets the client's needs. This section should include a plan to meet with the client towards the end of the project to discuss final design details and demonstrate a prototype, experimental test, and/or simulation results. Provide a relative time frame for this validation to occur (e.g. "mid-March" or "early-April"). Include a brief discussion on how client feedback will be captured, such as a

formal survey, interview, or observation notes of the client using the prototype. It may also include plans to solicit feedback from other stakeholders and/or potential users.

Section G. Results and Design Details

Use this section to highlight the major results of the design methodology described above including important analytical, computational, experimental, modeling, assembly, and testing results. This section should be one of the most substantial sections of the report showcasing all of the hard work and effort that went into the completion of the final design and delivery of the project deliverables. Show how the identified problem was solved.

Highlight the prominent features of the final design through analysis results, modeling, drawings, renderings, circuit schematics, instrumentation diagrams, flow and piping diagrams, etc. to show that the design functions as intended and meets all design objectives and constraints. Overview designs such as dataflow diagrams, process flow, swim lane diagrams, as well as presentation-layer designs (e.g. storyboards for front-ends) should be included here. Detailed designs such as database designs, software designs, procedure flowcharts, or pseudocode should be included here. Support computational and experimental results with key plots and figures. All supporting figures should be clearly labeled and annotated to highlight the most important points of the figure (i.e. explicitly point out what the reader should focus on or understand about the image).

Note that while all results should be used to help inform design decisions, not all results may be necessary to include in the main body of the report. Extraneous supporting results (e.g. graphs, data, design renderings, drawings, etc.) that are not necessary for presenting the fundamental findings can be placed in one or more appendices. Detailed documentation of each program module can be provided as appendix.

G.1 Modeling Results (example subsection)

G.2 Experimental Results (example subsection)

G.3 Prototyping and Testing Results (example subsection)

G.4. Final Design Details/Specifications (example subsection)

Note that while the design constraints and specifications may have provided minimum or maximum values, or ranges or values, that the design needed to meet, the final design specifications should be listed here showing that the required design values were met. A list of final design details can also be included demonstrate fulfillment of the design objectives.

Note: Preliminary results should be included in the Preliminary Design Report to show the progress made of the selected design concept to-date. This section should be updated for the Final Design Report to include documentation of all of the work that was completed on the project throughout the entirety of the academic year.

Section H. Societal Impacts of Design

In addition to technical design considerations, contemporary engineers must consider the broader impacts that their design choices have on the world around them. These impacts include the consideration of public health, safety, and welfare as well as the potential societal, political/regulatory, economic, environmental, global, and ethical impacts of the design. As appropriate for the project design, discuss how each of these considerations influenced design choices in separate subsections. How will the design change the way people interact with each other? What are the political implications of the design? Does the technology have the potential to impact or shift markets? Does the design have any positive or negative effects on the environment? Don't forget to consider unintended consequences such as process or manufacturing byproducts. What impacts might the design have on global markets and trade? Are there any ethical questions related to the design?

While it is hard to forecast the various impacts of a technology, it is important to consider these potential impacts throughout the engineering design process. When considered during the early stages of the design phase, consideration of these impacts can help determine design objectives, constraints, and specifications and help drive design choices that may mitigate any potential negative impacts or unintended consequences.

Note: A minimum of 4 of these design considerations, including the consideration of public health, safety, and welfare, are required for the Preliminary Design Report while a section for all considerations must be included in the final design report.

H.1 Public Health, Safety, and Welfare

Provide a list of all design safety features and provide a brief description of each. Discuss the potential effects the design may have on public health, safety, and welfare. References to the codes and standards previous provided and the organizations that produced them may be summarized or referenced here.

H.2 Societal Impacts

H.3 Political/Regulatory Impacts

H.4. Economic Impacts

H.5 Environmental Impacts

H.6 Global Impacts

H.7. Ethical Considerations

Section I. Cost Analysis

Provide a simple cost analysis of the project that includes a list of all expenditures related to the project. If an experimental test set-up or prototype was developed, provide a Bill of Materials that includes part numbers, vendor names, unit costs, quantity, total costs, delivery times, dates received, etc. Do not forget to include all manufacturing costs incurred throughout the completion of the project. If the design is expected to become a commercial product, provide a production cost estimate including fixed capital, raw materials, manufacturing (including tooling and/or casting), and labor costs to produce and package the device. Note that this type of detailed cost analysis may be listed as a project deliverable.

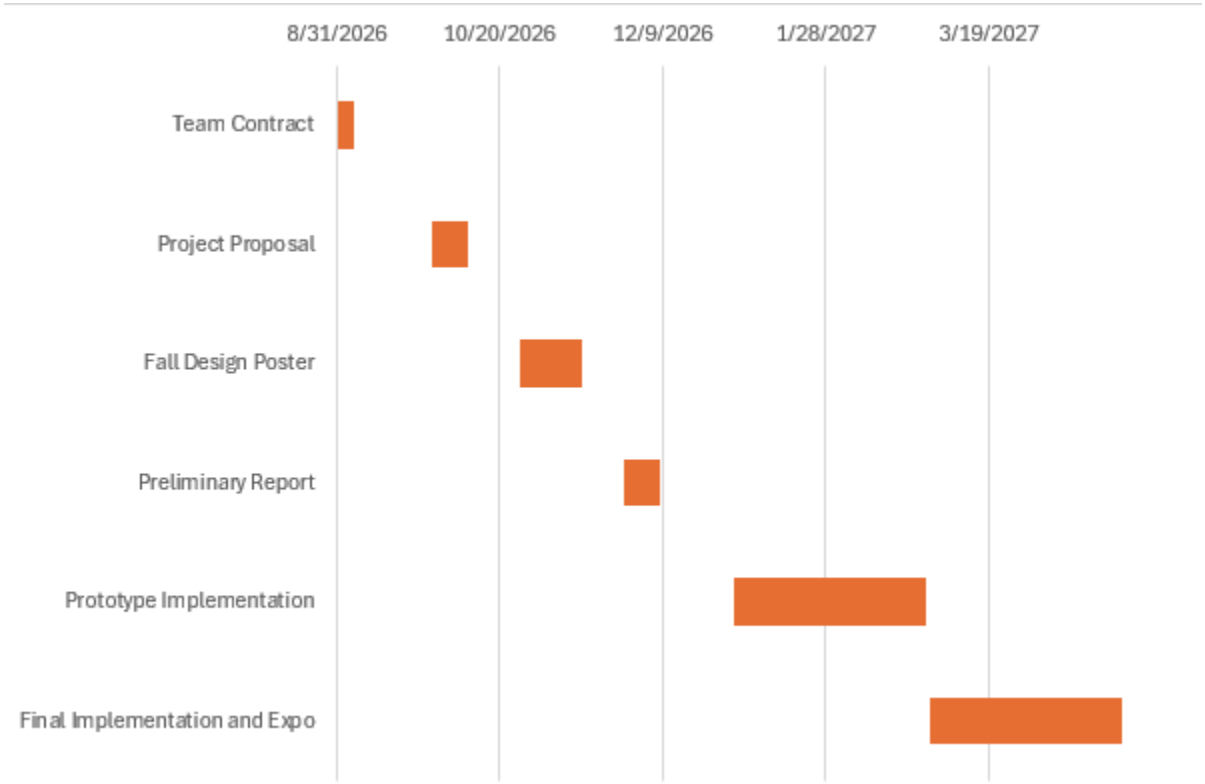
Note: The Preliminary Design Report should include all costs incurred to date. It is expected that this section will be expanded and updated between the preliminary and final design reports.

Section J. Conclusions and Recommendations

Use this section to summarize the story of how the design team arrived at the final design. Focus on the evolution of the design through the use of the engineering design process including lessons learned, obstacles overcome, and triumphs of the final design. Revisit the primary project goals and objectives. Provide a brief summary of the final design details and features paramount to the function of the design in meeting these goals and objectives.

A discussion may be included to discuss how the design could be further advanced or improved in the future. If applicable, summarize any questions or curiosities that the final results/design of this effort bring to mind or leave unanswered. If this project might continue on as a future (continuation) senior design project, detail the major milestones that have been completed to date and include any suggested testing plans, relevant machine drawings, electrical schematics, developed computer code, etc. All relevant information should be included in this section such that future researchers could pick up the project and advance the work in as seamless a manner as possible. Documents such as drawings, schematics, and codes could be referenced here and included in one or more appendix. If digital files are critical for future work, they should be saved on a thumb drive, external hard drive, cloud, etc. and left in the hands of the project advisor and/or client.

Appendix 1: Project Timeline



Appendix 2: Team Contract (i.e. Team Organization)

Step 1: Get to Know One Another. Gather Basic Information.

Task: This initial time together is important to form a strong team dynamic and get to know each other more as people outside of class time. Consider ways to develop positive working relationships with others, while remaining open and personal. Learn each other's strengths and discuss good/bad team experiences. This is also a good opportunity to start to better understand each other's communication and working styles.

Team Member Name	***Strengths each member bring to the group***	***Other Info***	***Contact Info***
Robert Smith	Report writing Database Development	Solid background in using SQL and databases in order to connect them to the frontend. Experienced in Python, Java, C.	smithre9@vcu.edu
Alycia Logan	Backend SQL Frontend		loganas2@vcu.edu
Adalberto Gomez	Python Software Backend		gomeza9@vcu.edu
Blake Cottrill	Python Frontend Software		cottrillbe@vcu.edu

Other Stakeholders	***Notes***	***Contact Info***
Thomas Gyeera		gyeeratw@vcu.edu

Step 2: Team Culture. Clarify the Group's Purpose and Culture Goals.

Task: Discuss how each team member wants to be treated to encourage them to make valuable contributions to the group and how each team member would like to feel recognized for their efforts. Discuss how the team will foster an environment where each team member feels they are accountable for their actions and the way they contribute to the project. These are your Culture Goals (left column). How do the students demonstrate these culture goals? These are your Actions (middle column). Finally, how do students deviate from the team's culture goals? What are ways that other team members can notice when that culture goal is no longer being honored in team dynamics? These are your Warning Signs (right column).

Resources: More information and an example Team Culture can be found in the Biodesign Student Guide "Intentional Teamwork" page (webpage | PDF)

Culture Goals	***Actions***	***Warning Signs***
Supportive	Support each other if a member is struggling with a goal. Cover for each other (within reason) if an event were to occur that would detriment a team member's workflow. Be overall respectful and considerate to create a good working environment.	If a team member starts to take advantage of the supportive culture to get out of work or abuses it by consistently pushing work onto the other team members.
Communication	Express goals and expectations for them openly. Constant communication of workflow in order to ensure that all team members are working towards the same goal and fulfilling each role efficiently.	When a team member consistently shows a lack of proper communication when working towards a goal or deadline. Not consistently showing up to team meetings and/or not responding to important group messages.
Trust	- Being accountable for your actions and taking the group members into consideration. - Feeling comfortable enough with group members to take the initiative and add value to the team.	- Refusing to engage with or help a team member while they are struggling with an assigned task (while they have the time and resources to do so). - Not doing work and disrespecting the

		shared deadline/goals.

Step 3: Time Commitments, Meeting Structure, and Communication

Task: Discuss the anticipated time commitments for the group project. Consider the following questions (don't answer these questions in the box below):

- What are reasonable time commitments for everyone to invest in this project?
- What other activities and commitments do group members have in their lives?
- How will we communicate with each other?
- When will we meet as a team? Where will we meet? How Often?
- Who will run the meetings? Will there be an assigned team leader or scribe? Does that position rotate or will same person take on that role for the duration of the project?

Required: How often you will meet with your faculty advisor, where you will meet, and how the meetings will be conducted. Who arranges these meetings?
See examples below.

Meeting Participants	***Frequency*** ***Dates and Times / Locations***	***Meeting Goals*** ***Responsible Party***
Blake Alycia Robert Adalberto	- Every Thursday @ 5pm - Zoom	Project Check-In, assigning tasks, and brainstorming
Blake Alycia Robert Adalberto	- Anytime - Groupme	General group concerns Any pressing matters
Robert Smith Thomas Gyeera	- Every Friday @ 1pm - Zoom	Advisor report, developing plan for coming weeks.

Step 4: Determine Individual Roles and Responsibilities

Task: As part of the Capstone Team experience, each member will take on a leadership role, ***in addition to*** contributing to the overall weekly action items for the project. Some common leadership roles for Capstone projects are listed below. Other roles may be assigned with approval of your faculty advisor as deemed fit for the project. For the entirety of the project, you should communicate progress to your advisor specifically with regard to your role.

- **Before meeting with your team**, take some time to ask yourself: what is my "natural" role in this group (strengths)? How can I use this experience to help me grow and develop more?
- **As a group,** discuss the various tasks needed for the project and role preferences. Then assign roles in the table on the next page. Try to create a team dynamic that is fair and equitable, while promoting the strengths of each member.

Communication Leaders

Suggested: Assign a team member to be the primary contact for faculty advisor. This person will schedule meetings, send updates, and ensure deliverables are met.

Common Leadership Roles for Capstone

1. **Project Manager:** Manages all tasks; develops overall schedule for project; writes agendas and runs meetings; reviews and monitors individual action items; creates an environment where team members are respected, take risks and feel safe expressing their ideas.
   Required: On Edusourced, under the Team tab, make sure that this student is assigned the Project Manager role. This is required so that Capstone program staff can easily identify a single contact person, especially for items like Purchasing and Receiving project supplies.
2. **Logistics Manager:** coordinates all internal and external interactions; lead in establishing contact within and outside of organization, following up on communication of commitments, obtaining information for the team; documents meeting minutes; manages facility and resource usage.
3. **Financial Manager:** researches/benchmarks technical purchases and acquisitions; conducts pricing analysis and budget justifications on proposed purchases; carries out team purchase requests; monitors team budget.
4. **Systems Engineer:** analyzes Client initial design specification and leads establishment of product specifications; monitors, coordinates and manages integration of sub-systems in the prototype; develops and recommends system architecture and manages product interfaces.
5. **Test Engineer:** oversees experimental design, test plan, procedures and data analysis; acquires data acquisition equipment and any necessary software; establishes test protocols and schedules; oversees statistical analysis of results; leads presentation of experimental finding and resulting recommendations.
6. **Manufacturing Engineer:** coordinates all fabrication required to meet final prototype requirements; oversees that all engineering drawings meet the requirements of machine shop or vendor; reviews designs to ensure design for manufacturing; determines realistic timing for fabrication and quality; develops schedule for all manufacturing.

Team Member	*Role(s)*	*Responsibilities*
Blake	Financial Manager	Takes care of all necessary purchases by researching and providing justification for each. Also needs to be able to monitor the budget for the project to ensure it is being maximized.
Alycia	Test Engineer	Designs and runs the test plan for the portal (functionality, security, performance), analyzes results, and presents findings and recommendations to guide development decisions.
Adalberto	Systems Engineer	Understand client specifications, create prototypes, build layout of project
Robert	Project & Logistics Manager	Run team meetings, determine general deadlines and schedules, and keep the team on track.

Step 5: Agree to the above team contract

Team Member: Alycia Logan *Signature:* 

Team Member: Robert Smith *Signature:* 

Team Member: Adalberto Gomez *Signature:* 

Team Member: Blake Cottrill *Signature:* 

References

[1] VCU Writing Center. (2021, September 8). *APA Citation: A guide to formatting in APA style.* Retrieved September 2, 2024. https://writing.vcu.edu/student-resources/apa-citations/

[2] *Modules - FHIR V5.0.0*, hl7.org/fhir/modules.html. Accessed 10 Oct. 2026.